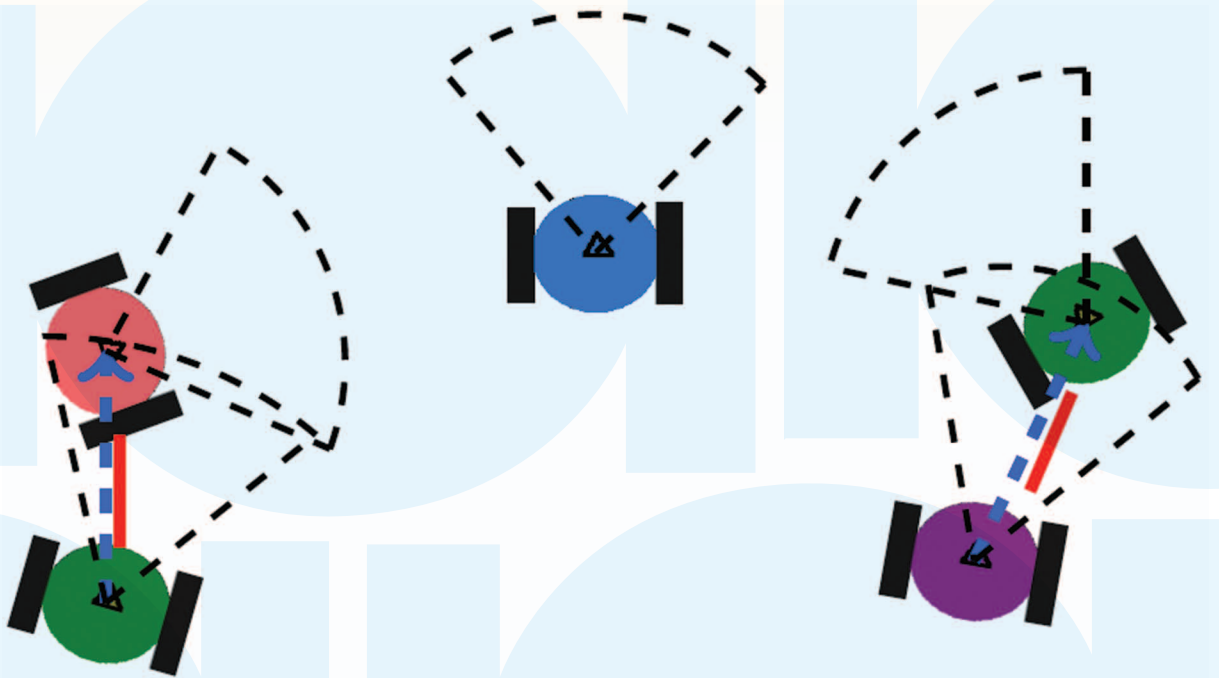


Technological advances in ad hoc networking and the miniaturization of electromechanical systems are making possible the use of large numbers of mobile agents (for example, mobile robots, human agents, and unmanned vehicles) to perform surveillance, search and rescue, transport, and delivery tasks in aerial, underwater, space, and land environments. However, the successful execution of such tasks often hinges upon accurate position information, which is needed in lower-level locomotion and path-planning algorithms. Common techniques for the localization of mobile robots are the classical preinstalled beacon-based localization algorithms [1], fixed feature-based simultaneous localization and mapping (SLAM) algorithms [2], and Global Positioning System (GPS) navigation [3]; see Figure 1 for further details. However, these localization techniques work based on assumptions such as the existence of distinct and static features that can be revisited



Cooperative Localization for Mobile Agents

**A RECURSIVE DECENTRALIZED ALGORITHM
BASED ON KALMAN-FILTER DECOUPLING**

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often or line of sight to GPS satellites, which may not be feasible for operations such as search and rescue [4], [5], environment monitoring [6], [7], and oceanic exploration [8]. In the case of GPS navigation, there is also a current concern about signal jamming for outdoor navigation, especially for unmanned aerial vehicle coordination and control. Instead, cooperative localization (CL) is emerging as an alternative localization technique that can be employed in such scenarios.

In CL, a group of mobile agents with processing and communication capabilities use relative measurements with respect to each other (no reliance on external features) as a feedback signal to *jointly estimate* the poses (both position and orientation) of all team members, which results in increased accuracy for the entire team. The principle behind CL is similar to that of differential GPS, by which inaccurate GPS measurements are corrected with range measurements from stations with known locations. However, CL goes beyond differential GPS since it assumes no fixed stations or landmarks or continuous access to GPS measurements (even if they are inaccurate).

The particular appeal of CL relies on the fact that sporadic access to accurate localization information by a particular robot results in a net benefit for the rest of the team. This is possible thanks to the coupling that is created through the state-estimation process. Moreover, the use of other robots as landmarks also eliminates the potential problems on data association and perception because the robots can be equipped with distinctive features or markers that are easily recognizable by other robots. Another nice feature of CL is its cost effectiveness, since it does not require extra hardware beyond the operational components

normally used in cooperative robotic tasks. In such situations, agents are normally equipped with unique identifiers and sensors that enable them to identify and locate other group members. To achieve coordination, these agents often broadcast their status information to one another. In addition, given the affordability and wide availability of communication devices, CL has also emerged as an augmentation system to compensate for poor odometric measurements, and/or noisy and distorted measurements from other sensor suites, such as onboard inertial measurement units; see, for example, [9].

The idea of exploiting relative robot-to-robot measurements for localization can be traced back to [10], where members of a mobile robotic team were divided into two groups, which took turns remaining stationary as landmarks for the others. Later, [11] removed the necessity of some robots to be stationary and also introduced the term CL to refer to this localization technique. Since then, many CL algorithms, using various estimation strategies such as extended Kalman filters (EKF) [12], maximum likelihood [13], maximum a posteriori (MAP) [14], and particle filters [15]–[17], have been developed. CL techniques to handle system and measurement models with non-Gaussian noises are also discussed in [18] and [19].

Although CL is a very attractive concept for multirobot localization because of its lack of dependency on environmental features or GPS information, it presents new implementation challenges associated with achieving acceptable communication, memory, and processing costs. CL is a joint estimation process that results in highly coupled pose estimation for the entire robotic team. These couplings/cross correlations are created due to the relative measurement

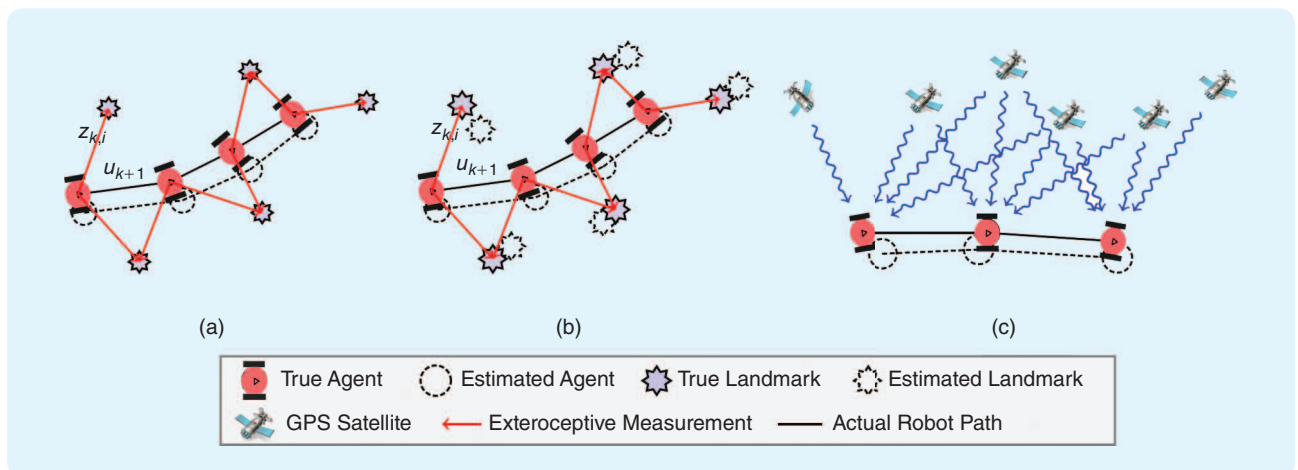


FIGURE 1 A schematic representation of common probabilistic localization techniques for mobile platforms: (a) beacon-based localization, (b) simultaneous localization and mapping (SLAM) [35], and (c) Global Positioning System (GPS) navigation. In beacon-based localization, the map of the area is known and there are preinstalled beacons or landmarks with known locations and identities. By taking relative measurements with respect to these landmarks, the mobile agents can improve their localization accuracy. For operations where a priori knowledge about the environment is not available, but the environment contains fixed and distinguishable features that agents can measure, SLAM is normally used to localize the mobile agents. SLAM is a process by which a mobile agent can build a map of an environment and at the same time use this map to deduce its location. On the other hand, GPS navigation provides location and time information in all weather conditions, anywhere on or near the earth but it requires a clear line of sight to at least four GPS satellites.

This article presents a brief review on cooperative localization as a strategy to increase the localization accuracy of a team of mobile agents with communication capabilities.

updates. Accounting for these coupling/cross correlations is crucial for both filter consistency and also for propagating the benefit of a robot-to-robot measurement update to the entire group. In the section “Cooperative Localization via the Extended Kalman Filter,” these features are highlighted through detailed examination of equations of a centralized EKF CL and a simulation demonstration.

A centralized implementation of CL is the most straightforward mechanism to keep an accurate account of the couplings and, as a result, to obtain more accurate solutions. In a centralized scheme, *at every time step*, a single device, either a leader robot or a fusion center (FC), gathers and processes information from the entire team. Then, the FC broadcasts back the estimated location results to each robot (see, for example, [12] and [20]). But such a central operation is highly energy consuming since it incurs a high processing cost on the FC and a high communication cost on both

the FC and each robotic team member. Moreover, central operations lack robustness due to existence of single failure point. This energy inefficiency and lack of robustness make the centralized implementation less preferable.

A major challenge in developing decentralized CL (DCL) algorithms is how to maintain a precise account of cross correlations and couplings between the agents’ estimates without invoking *all-to-all* communication at each time step. The design and analysis of DCL algorithms that maintain the consistency of the estimation process while maintaining reasonable communication and computation costs have been the subject of extensive research since the conception of the CL idea. The section “Decentralized Cooperative Localization: How to Account for Intrinsic Correlations in Cooperative Localization,” gives an overview of some of the DCL algorithms in the literature, with a special focus on how these algorithms maintain or

Further Reading

A performance analysis of an EKF CL for a team of homogeneous robots moving on a flat terrain, with the same level of uncertainty in their proprioceptive measurements and exteroceptive sensors that measure relative pose, is provided in [S1] and [S2]. Interestingly, [S1] shows that the rate of uncertainty growth decreases as the size of the robot team increases but is subject to the law of diminishing returns. Moreover, [S2] shows that the upper bound on the rate of uncertainty growth is independent of the accuracy or the frequency of the robot-to-robot measurements. The consistency of the EKF CL from the perspective of observability is studied in [S3], which analytically shows that the error-state system model employed in the standard EKF CL always has an observable subspace of higher dimension than that of the actual nonlinear CL system. This results in an unjustified reduction of the EKF covariance estimates in directions of the state space where no information is available and thus leads to inconsistency. To address this problem, [S3] adopts an observability-based methodology for designing consistent estimators in which the linearization points are selected to ensure a linearized system model with an observable subspace of the correct dimension. More results on observability analysis of CL can be found in [22], [S4], and [S5]. The use of an observability analysis to explicitly design an active local path-planning algorithm for unmanned aerial vehicles implementing a bearing-only CL is discussed in [S6]. The necessity for an initialization procedure for

CL is discussed in [S7], and it is shown that, because of system nonlinearities and the periodicity of the orientation, initialization errors can lead to erroneous results in covariance-based filters. An initialization procedure for the state estimate in a CL scenario based on ranging and dead reckoning is studied in [S8].

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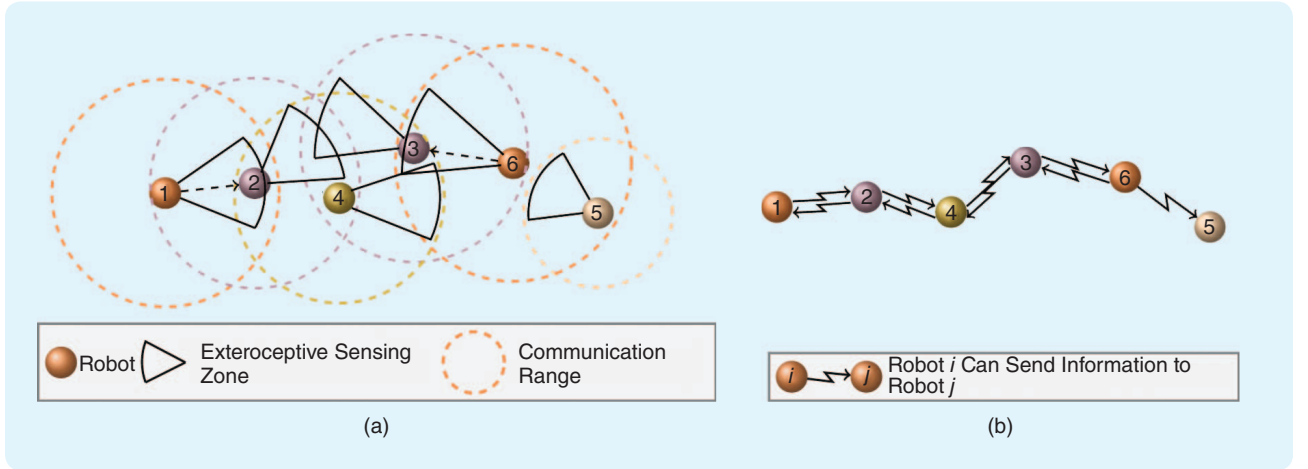


FIGURE 2 A multihop communication scenario for the multirobot team. Plot (a) shows the communication and measurement ranges. Here, robots 1 and 6 make relative measurements, respectively, of robots 2 and 3. Plot (b) shows the communication graph generated using the communication ranges given in (a). Here the robot at the head of an arrow can send information to the robot at the tip of the arrow. As this graph shows, each of robots 1 and 6 can pass communication message to the entire team via a multihop strategy.

account for intrinsic correlations of CL strategy. Subsequently, the section “The Interim Master DCL Algorithm: A Tightly Coupled DCL Strategy Based on Kalman-Filter Decoupling,” provides an example of a DCL algorithm that maintains the exact account of cross covariances between robotic team-member pose estimates without resorting to all-to-all communication. A preliminary version of this example DCL strategy appeared in [21]. Those interested in technical analysis and details beyond decentralization for CL can find a brief literature guide in “Further Reading.”

Notation

Before proceeding further, the notations are introduced. $\mathbb{M}_n$, $\mathbf{0}_{n \times m}$ (when $m = 1$, $\mathbf{0}_n$ is used), and $\mathbf{I}_n$, respectively, denote the set of real positive-definite matrices of dimension $n \times n$, the zero matrix of dimension $n \times m$, and the identity matrix of dimension $n \times n$. $\mathbf{A}^\top$ represents the transpose of matrix $\mathbf{A} \in \mathbb{R}^{n \times m}$. The block diagonal matrix of set of matrices $\mathbf{A}_1, \dots, \mathbf{A}_N$ is $\text{Diag}(\mathbf{A}_1, \dots, \mathbf{A}_N)$. For finite sets V_1 and V_2 , $V_1 \setminus V_2$ is the set of elements in V_1 , but not in V_2 . For a finite set V its cardinality is represented by $|V|$. In a team of N agents, the local variables associated with agent i are distinguished by the superscript i , for example, $\mathbf{x}^i$ is the state of agent i , $\hat{\mathbf{x}}^i$ is its state estimate, and $\mathbf{P}^i$ is the covariance matrix of its state estimate. In this article, the term *cross covariance* refers to the correlation terms between two agents in the covariance matrix of the entire team. The cross covariance of the state vectors of agents i and j is $\mathbf{P}_{ij}$. The propagated and updated variables, say $\hat{\mathbf{x}}^i$, at time step k are denoted by $\hat{\mathbf{x}}^{i-}(k)$ and $\hat{\mathbf{x}}^{i+}(k)$, respectively. The time-step argument of the variables as well as matrix dimensions are dropped whenever they are clear from the context. In a network of N agents, $\mathbf{p} = (\mathbf{p}^1, \dots, \mathbf{p}^N) \in \mathbb{R}^d$, $d = \sum_{i=1}^N n^i$ is the aggregated vector of local vectors $\mathbf{p}^i \in \mathbb{R}^{n^i}$.

COOPERATIVE LOCALIZATION VIA THE EXTENDED KALMAN FILTER

This section provides an overview of a CL strategy that employs an EKF following [22]. By a close examination of this algorithm, it is possible to explain why accounting for the intrinsic cross correlations in CL is both crucial for filter consistency and a key to spreading the benefit of an update of a relative robot-to-robot measurement to the entire team. This section also discusses the communication cost of implementing this algorithm in a centralized manner.

Consider a group of N mobile agents with communication, processing, and measurement capabilities. Depending on the adopted CL algorithm, communication can be in 1) a bidirectional manner with an FC, 2) a single broadcast to the entire team, or 3) multihop fashion of every agent rebroadcasting every received message intended to reach the entire team; see Figure 2. Each agent has a detectable unique identifier, which, without loss of generality, is assumed to be a unique integer belonging to the set $\mathcal{V} = \{1, \dots, N\}$. Using a set of proprioceptive sensors, every agent $i \in \mathcal{V}$ measures its self-motion, for example, by compass readings and/or wheel encoders, and uses it to propagate its equations of motion

$$\mathbf{x}^i(k+1) = \mathbf{f}^i(\mathbf{x}^i(k), \mathbf{u}^i(k)) + \mathbf{g}^i(\mathbf{x}^i(k))\boldsymbol{\eta}^i(k), \quad (1)$$

where $\mathbf{x}^i \in \mathbb{R}^{n^i}$, $\mathbf{u}^i \in \mathbb{R}^{m^i}$, and $\boldsymbol{\eta}^i \in \mathbb{R}^{p^i}$ are, respectively, the state vector, the input vector, and the process noise vector of agent i . Here, $\mathbf{f}^i(\mathbf{x}^i, \mathbf{u}^i)$ and $\mathbf{g}^i(\mathbf{x}^i)$ are, respectively, the system function and process noise coefficient function of the agent $i \in \mathcal{V}$. The state vector of each agent can be composed of variables that describe the robot's global pose in the world (for example, latitude, longitude, direction) as well as other variables potentially needed to model the robots dynamics (for example, steering angle and actuation

ALGORITHM 1 Extended Kalman filter cooperative localization (centralized).

Require: Initialization ($k = 0$): For $i \in \mathcal{V}$, the algorithm is initialized at

$$\hat{\mathbf{x}}^{i+}(0) \in \mathbb{R}^{n^i}, \mathbf{P}^{i+}(0) \in \mathbb{M}^{n^i}, \mathbf{P}_{ij}^+(0) = \mathbf{0}_{n^i \times n^j}, j \in \mathcal{V} \setminus \{i\}.$$

Iteration k

1: Propagation: for $i \in \mathcal{V}$, the propagation equations are

$$\hat{\mathbf{x}}^{i-}(k+1) = \mathbf{f}^i(\hat{\mathbf{x}}^{i+}(k), \mathbf{u}^i(k)), \quad (\text{S1a})$$

$$\mathbf{P}^{i-}(k+1) = \mathbf{F}^i(k) \mathbf{P}^{i+}(k) \mathbf{F}^{i\top}(k) + \mathbf{G}^i(k) \mathbf{Q}^i(k) \mathbf{G}^{i\top}(k), \quad (\text{S1b})$$

$$\mathbf{P}_{ij}^-(k+1) = \mathbf{F}^i(k) \mathbf{P}_{ij}^+(k) \mathbf{F}^j(k)^\top, \quad j \in \mathcal{V} \setminus \{i\}. \quad (\text{S1c})$$

2: Update: While there are no relative measurements no update happens, that is,

$$\hat{\mathbf{x}}^+(k+1) = \hat{\mathbf{x}}^-(k+1), \mathbf{P}^+(k+1) = \mathbf{P}^-(k+1).$$

When there is a relative measurement at time step $k+1$, for example robot a makes a relative measurement of robot b , the update proceeds as below. The innovation of the relative measurement and its covariance are, respectively,

$$\mathbf{r}^a = \mathbf{z}_{ab} - \mathbf{h}_{ab}(\hat{\mathbf{x}}^{a-}(k+1), \hat{\mathbf{x}}^{b-}(k+1)),$$

and

$$\begin{aligned} \mathbf{S}_{ab} = & \mathbf{R}^a(k+1) + \tilde{\mathbf{H}}_a(k+1) \mathbf{P}^{a-}(k+1) \tilde{\mathbf{H}}_a^\top(k+1) \\ & + \tilde{\mathbf{H}}_b(k+1) \mathbf{P}^{b-}(k+1) \tilde{\mathbf{H}}_b^\top(k+1) \\ & - \tilde{\mathbf{H}}_b(k+1) \mathbf{P}_{ba}^-(k+1) \tilde{\mathbf{H}}_a^\top(k+1) - \tilde{\mathbf{H}}_a(k+1) \mathbf{P}_{ab}^-(k+1) \tilde{\mathbf{H}}_b^\top(k+1). \end{aligned} \quad (\text{S2})$$

The estimation updates for the centralized EKF are

$$\hat{\mathbf{x}}^{i+}(k+1) = \hat{\mathbf{x}}^{i-}(k+1) + \mathbf{K}_i(k+1) \mathbf{r}^a(k+1), \quad (\text{S3a})$$

$$\mathbf{P}^{i+}(k+1) = \mathbf{P}^{i-}(k+1) - \mathbf{K}_i(k+1) \mathbf{S}_{ab}(k+1) \mathbf{K}_i^\top(k+1), \quad (\text{S3b})$$

$$\mathbf{P}_{ij}^+(k+1) = \mathbf{P}_{ij}^-(k+1) - \mathbf{K}_i(k+1) \mathbf{S}_{ab}(k+1) \mathbf{K}_j^\top(k+1), \quad (\text{S3c})$$

where $i \in \mathcal{V}$, $j \in \mathcal{V} \setminus \{i\}$ and

$$\mathbf{K}_i = (\mathbf{P}_{ib}^-(k+1) \tilde{\mathbf{H}}_b^\top - \mathbf{P}_{ia}^-(k+1) \tilde{\mathbf{H}}_a^\top) \mathbf{S}_{ab}^{-1}. \quad (\text{S4})$$

3: $k \leftarrow k+1$

the measurement model and $\mathbf{v}^i$ is the measurement noise of agent $i \in \mathcal{V}$. Relative-measurement feedback, as shown below, can help the agents improve their localization accuracy, although the overall uncertainty cannot be bounded; see [22]. The tracking performance can be improved significantly if agents have occasional absolute positioning information, for example, via GPS or relative measurements taken from a fixed landmark with a priori known absolute location. Any absolute pose measurement by an agent $i \in \mathcal{V}$, for example, through intermittent GPS access, is modeled by $\mathbf{z}_{ii}(k+1) = \mathbf{h}_{ii}(\mathbf{x}^i(k)) + \mathbf{v}^i(k)$. The agents can obtain concurrent exteroceptive absolute and relative measurements.

In the technical developments below, all the process noises $\boldsymbol{\eta}^i$ and the measurement noise $\mathbf{v}^i$, $i \in \mathcal{V}$, are assumed to be independent zero-mean white Gaussian processes with, respectively, known positive-definite variances $\mathbf{Q}^i(k) = E[\boldsymbol{\eta}^i(k) \boldsymbol{\eta}^{i\top}(k)]$, $\mathbf{R}^i(k) = E[\mathbf{v}^i(k) \mathbf{v}^{i\top}(k)]$, and $\tilde{\mathbf{R}}^i(k) = E[\tilde{\mathbf{v}}^i(k) \tilde{\mathbf{v}}^{i\top}(k)]$. Moreover, all the sensor noises are white and mutually uncorrelated and all sensor measurements are synchro-

nized. Then, the centralized EKF CL algorithm is a straightforward application of EKF over the collective motion model of the robotic team (2) and measurement model (3). The propagation stage of this algorithm is

$$\mathbf{x}(k+1) = \mathbf{f}(\mathbf{x}(k), \mathbf{u}(k)) + \mathbf{g}(\mathbf{x}(k)) \boldsymbol{\eta}(k), \quad (2)$$

where $\mathbf{x} = (\mathbf{x}^1, \dots, \mathbf{x}^N)$, $\mathbf{u} = (\mathbf{u}^1, \dots, \mathbf{u}^N)$, $\boldsymbol{\eta} = (\boldsymbol{\eta}^1, \dots, \boldsymbol{\eta}^N)$, $\mathbf{f}(\mathbf{x}, \mathbf{u}) = (\mathbf{f}^1(\mathbf{x}^1, \mathbf{u}^1), \dots, \mathbf{f}^N(\mathbf{x}^N, \mathbf{u}^N))$, and $\mathbf{g}(\mathbf{x}) = \text{Diag}(\mathbf{g}^1(\mathbf{x}^1), \dots, \mathbf{g}^N(\mathbf{x}^N))$.

If each agent only relies on propagating its equation of motion in (1) using self-motion measurements, the state estimate error drifts due to the noise term $\boldsymbol{\eta}^i(k)$, with a standard deviation that grows in time without bound. To reduce the growth rate of this estimation error, a CL strategy can be employed. Thus, let every agent $i \in \mathcal{V}$ also carry exteroceptive sensors to monitor the environment to detect, uniquely, the other agents $j \in \mathcal{V}$ in the team and take relative measurements

$$\mathbf{z}_{ij}(k+1) = \mathbf{h}_{ij}(\mathbf{x}^i(k), \mathbf{x}^j(k)) + \mathbf{v}^i(k), \quad (3)$$

where $\mathbf{z}_{ij} \in \mathbb{R}^{n_z^i}$. These measurements can include relative pose, relative range, and relative bearing. Here, $\mathbf{h}_{ij}(\mathbf{x}^i, \mathbf{x}^j)$ is

where $\mathbf{F} = \text{Diag}(\mathbf{F}^1, \dots, \mathbf{F}^N)$, $\mathbf{G} = \text{Diag}(\mathbf{G}^1, \dots, \mathbf{G}^N)$, and $\mathbf{Q} = \text{Diag}(\mathbf{Q}^1, \dots, \mathbf{Q}^N)$, with, for all $i \in \mathcal{V}$, $\mathbf{F}^i = (\partial/\partial \mathbf{x}^i) \mathbf{f}(\hat{\mathbf{x}}^{i+}(k), \mathbf{u}^i(k))$, and $\mathbf{G}^i = (\partial/\partial \mathbf{x}^i) \mathbf{g}(\hat{\mathbf{x}}^{i+}(k))$.

$$\hat{\mathbf{x}}^-(k+1) = \mathbf{f}(\hat{\mathbf{x}}^+(k), \mathbf{u}(k)), \quad (4a)$$

$$\mathbf{P}^-(k+1) = \mathbf{F}(k) \mathbf{P}^+(k) \mathbf{F}^\top(k) + \mathbf{G}(k) \mathbf{Q}(k) \mathbf{G}^\top(k), \quad (4b)$$

If there exists a relative measurement in the network at some given time $k+1$, say robot a takes relative measurement from robot b , the states are updated as follows. The innovation of the relative measurement and its covariance are, respectively,

$$\mathbf{r}^a = \mathbf{z}_{ab} - \mathbf{h}_{ab}(\hat{\mathbf{x}}^{a-}(k+1), \hat{\mathbf{x}}^{b-}(k+1)), \quad (5a)$$

$$\mathbf{S}_{ab} = \mathbf{H}_{ab}(k+1) \mathbf{P}^-(k+1) \mathbf{H}_{ab}^\top(k+1) + \mathbf{R}^a(k+1), \quad (5b)$$

where (without loss of generality $a < b$)

$$\begin{aligned}\mathbf{H}_{ab}(k) &= [\mathbf{0} \quad \cdots \quad -\tilde{\mathbf{H}}_a^a(k) \quad \mathbf{0} \quad \cdots \quad \tilde{\mathbf{H}}_b^b(k) \quad \mathbf{0} \quad \cdots], \\ \tilde{\mathbf{H}}_a^a(k) &= -\frac{\partial}{\partial \mathbf{x}^a} \mathbf{h}_{ab}(\hat{\mathbf{x}}^{a-}(k), \hat{\mathbf{x}}^{b-}(k)), \\ \tilde{\mathbf{H}}_b^b(k) &= \frac{\partial}{\partial \mathbf{x}^b} \mathbf{h}_{ab}(\hat{\mathbf{x}}^{a-}(k), \hat{\mathbf{x}}^{b-}(k)).\end{aligned}\quad (6)$$

An absolute measurement by a robot $a \in \mathcal{V}$ can be processed similarly, except that in (6), $\tilde{\mathbf{H}}_b$ becomes zero, while in (5), the index b should be replaced by a and $\mathbf{R}^a(k+1)$ should be replaced by $\tilde{\mathbf{R}}^a(k+1)$. Then, the Kalman filter gain is

$$\mathbf{K}(k+1) = \mathbf{P}^-(k+1) \mathbf{H}_{ab}(k+1)^\top \mathbf{S}_{ab}^{-1}.$$

Finally, the collective pose update and covariance update equations for the robotic team are, respectively,

$$\hat{\mathbf{x}}^+(k+1) = \hat{\mathbf{x}}^-(k+1) + \mathbf{K}(k+1) \mathbf{r}^a, \quad (7a)$$

$$\mathbf{P}^+(k+1) = \mathbf{P}^-(k+1) - \mathbf{K}(k+1) \mathbf{S}_{ab} \mathbf{K}(k+1)^\top. \quad (7b)$$

Because $\mathbf{K}(k+1) \mathbf{S}_{ab} \mathbf{K}(k+1)^\top$ is a positive semidefinite term, the update equation (7b) shows that any relative measurement update results in a reduction of the estimation uncertainty.

To explore the relationship among the estimation equations of each robot, the aforementioned collective form of the EKF CL is expressed in terms of its agent-wise components in Algorithm 1. Here, the Kalman filter gain is partitioned into $\mathbf{K} = [\mathbf{K}_1^\top, \dots, \mathbf{K}_N^\top]^\top$, where $\mathbf{K}_i \in \mathbb{R}^{n' \times n_i^z}$ is the portion of the Kalman gain used to update the pose estimate of the agent $i \in \mathcal{V}$. To process multiple synchronized measurements, *sequential updating* is employed (consult, for example, [23, Ch. 3] and [24]).

Algorithm 1 showcases the role of past correlations in a CL strategy. First, observe that, despite having decoupled equations of motion, the source of the coupling in the

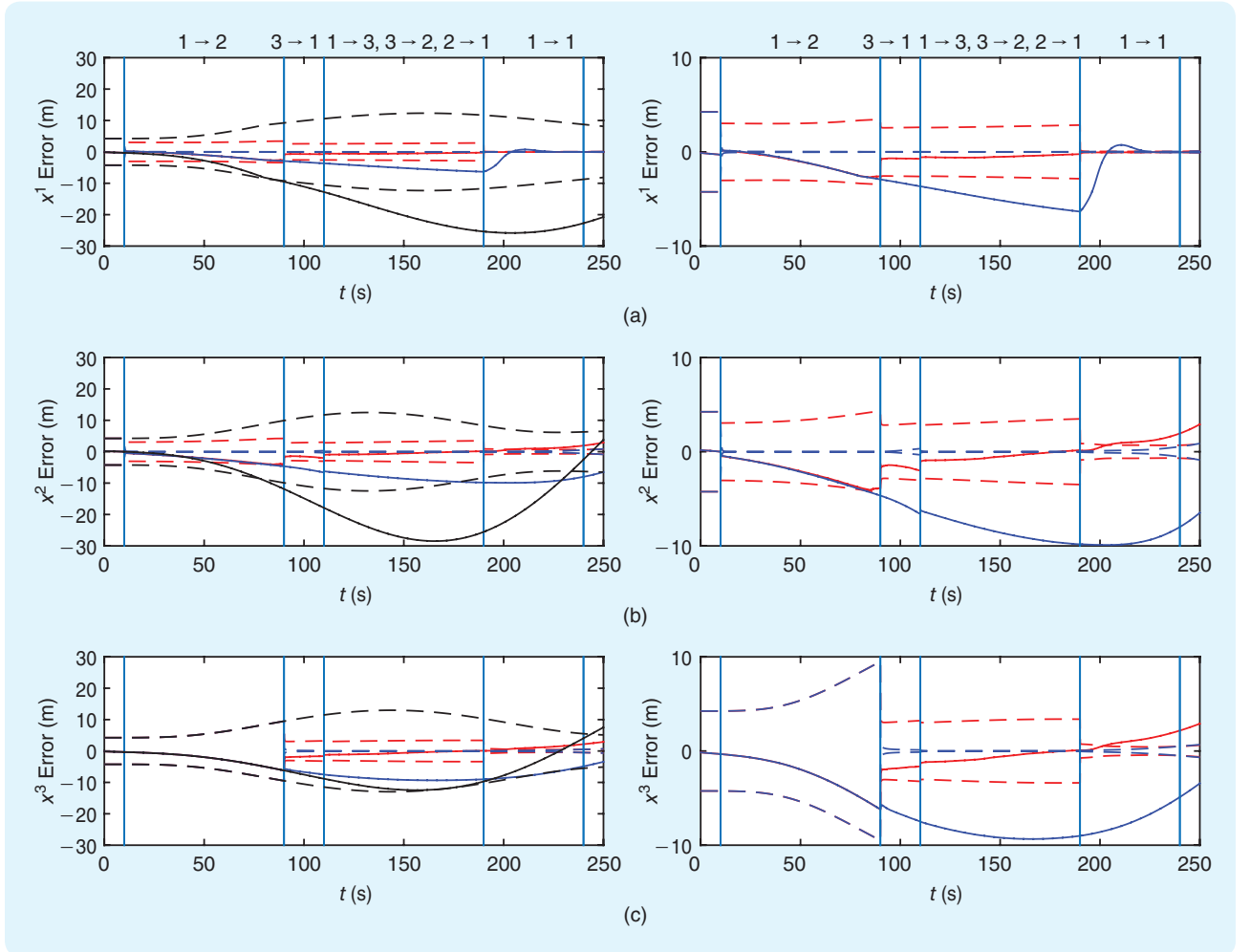


FIGURE 3 Estimation error (solid line) and 3σ error bounds (dashed lines) in the x -coordinate variable for three robots moving on a flat terrain when they (a) only propagate their equations of motion using self-motion measurements (black plots), (b) employ cooperative localization ignoring past correlations between the estimates of the robots (blue plots), and (c) employ cooperative localization with accurate account of past correlations (red plots). The figures in the right column are the same figures as on the left where the localization case (a) is removed for clearer demonstration of cases (b) and (c). Here, $a \rightarrow b$ over the time interval marked by two vertical blue lines indicates that robot a has taken a relative measurement with respect to robot b at that time interval. The symbol $a \rightarrow a$ means that robot a obtains an absolute measurement.

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to accurate localization information by a particular robot
results in a net benefit for the rest of the team.**

propagation phase is the cross-covariance equation (S1c). Upon incidence of a relative measurement between agents a and b , the cross-covariance term between robots a and b becomes nonzero, and its evolution in time requires the information of these two agents. Thus, these two agents have to either communicate with each other all the time or a central operator has to take over the propagation stage. As the number of relative measurements grow, more nonzero cross-covariance terms are created. As a result, the communication cost to perform the propagation grows, requiring the data exchange all the time with either an FC or all-to-all agent communications, even when there is no relative measurement in the network. The update equations (S3) are also coupled and their calculations need, in principle, an FC. The next observation regarding the role of the cross-covariance terms can be deduced from studying the Kalman-gain equation (S4). As this equation shows, when an agent a takes a relative measurement from agent b , any agent whose pose estimate is correlated with either of agents a and b in the past, (that is, $\mathbf{P}_{ib}^-(k+1)$ and/or $\mathbf{P}_{ia}^-(k+1)$ are nonzero) has a nonzero Kalman gain and, as a result, the agent benefits from this measurement update. The same is true in the case of an absolute measurement taken by a robot a .

The following simple simulation study demonstrates the significance of maintaining an accurate account of cross-covariance terms between the state estimates of the team members. This simulation study involves a team of three mobile robots moving on a flat terrain whose equa-

tions of motion in a fixed reference frame, for $i \in \{1, 2, 3\}$, are modeled as

$$\begin{aligned}x^i(k+1) &= x^i(k) + V^i(k) \cos(\phi(k)) \delta t, \\y^i(k+1) &= y^i(k) + V^i(k) \sin(\phi(k)) \delta t, \\ \phi^i(k+1) &= \phi^i(k) + \omega(k) \delta t,\end{aligned}$$

where $V^i(k)$ and $\omega^i(k)$ are true linear and rotational velocities of robot i at time k and δt is the step size. Here, the pose vector of each robot is $\mathbf{x}^i = [x^i, y^i, \phi^i]^\top$. Every robot uses odometry sensors to measure its linear $V_m^i(k) = V^i(k) + \eta_V^i(k)$ and rotational $\omega_m^i(k) = \omega^i(k) + \eta_\omega^i(k)$ velocities, where η_V^i and η_ω^i are their respective contaminating measurement noises. The standard deviation of $\eta_V^i(k)$, $i \in \{1, 2, 3\}$, is $0.1V^i(k)$, while the standard deviation of η_ω^i is $1^\circ/\text{s}$ for robot 1 and robot 2, and $0.5^\circ/\text{s}$ for robot 3. Robots $\{1, 2, 3\}$ can take relative pose measurements from one another. This relative measurements are corrupted by additive Gaussian measurement noise with standard deviations of, respectively, (0.05 m, 0.05 m, $1^\circ/\text{s}$), (0.05 m, 0.05 m, $2^\circ/\text{s}$), (0.07 m, 0.07 m, $1.5^\circ/\text{s}$) for robots 1, 2, and 3. Here, it is also assumed that robot 1 can obtain absolute position measurements with a standard deviation of (0.1 m; 0.1 m) for the measurement noise. Figure 3 demonstrates the x -coordinate estimation error (solid line) and the 3σ error bound (dashed lines) of these robots when they 1) only propagate their equations of motion using self-motion measurements (black plots), 2) employ an EKF CL ignoring past correlations between the estimates of the robots (blue plots), and 3) employ an EKF CL

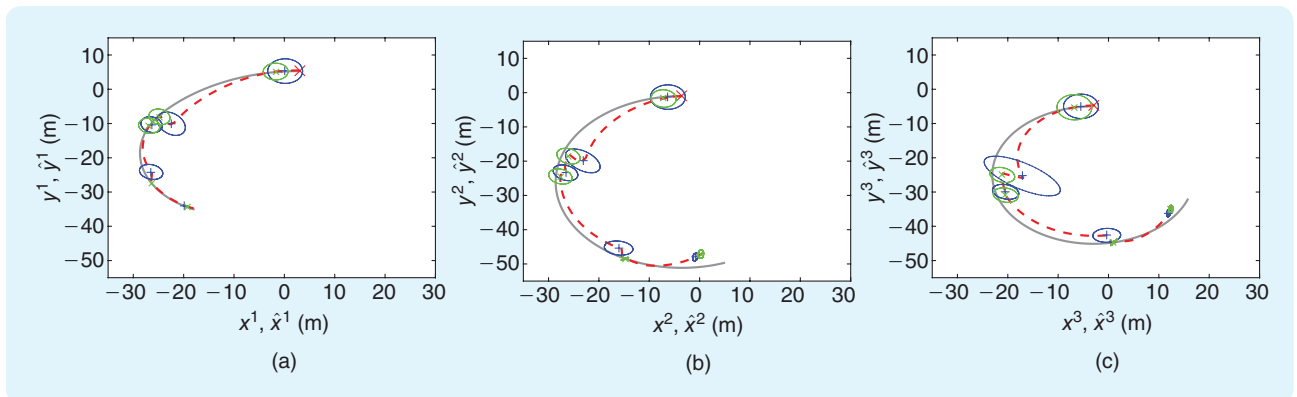


FIGURE 4 The trajectories of the robots for the simulation study of Figure 3. Here, the gray curve is the ground truth. The red curve is the estimates of the trajectory by implementing an extended Kalman filter cooperative localization. The blue (respectively, green) ellipses show the 95% uncertainty regions for the estimations at 2 s before (respectively, after) any change in the measurement scenario (see Figure 3). (a) Robot 1. (b) Robot 2. (c) Robot 3.

[29] and [30]) to fuse the newly acquired pose estimate with its own current estimate to increase its estimation accuracy. Covariance intersection for DCL is also used in [26] for the localization of a group of space vehicles communicating over a fixed ring topology. Here, each vehicle propagates a model of the equation of motion of the entire team, and, at the time of relative pose measurements, the vehicle fuses its estimate of the team states and of its landmark vehicle using covariance intersection. Another example of the use of split covariance intersection is given in [27] for intelligent transportation vehicles localization. A common past-invariant, ensemble Kalman pose estimation filter is used in another loosely coupled CL approach for intelligent vehicles [28]. That algorithm is very similar to the decentralized covariance intersection data fusion method described above, with the main difference being that it operates with ensembles instead of with means and covariances. Even though the covariance intersection method produces consistent estimates for a loosely coupled DCL strategy, this method is known to produce overly conservative estimates. Overall, the loosely coupled algorithms have the advantage of not imposing any particular connectivity condition on the team. However, they are conservative by nature because they do not enable other agents in the network to fully benefit from measurement updates.

In the tightly coupled DCL methodology, the goal is to exploit the network of correlations created across the team (see Figure 5) so that the benefit of the update can be extended beyond the agents involved in a given relative measurement. However, this advantage comes at a potentially higher computational, storage, and/or communication cost. The dominant trend in developing DCL algorithms in this way is to distribute the computation of components of a centralized algorithm among team members. Some of the examples for this class of DCL are given in [22], [14], and [31]–[33]. In a straightforward fashion, decentralization can be conducted as a multicentralized CL, wherein each agent broadcasts its own information to the entire team. Then, every agent can calculate and reproduce the centralized pose estimates acting as an FC [31]. Besides a high processing cost for each agent, this scheme requires all-to-all agent communication at the time of each information exchange. A DCL algorithm distributing computations of an EKF centralized CL algorithm is proposed in [22]. To decentralize the cross-covariance propagation, [22] uses a singular-value decomposition to split each cross-covariance term between the corresponding two agents. Then, each agent propagates its portion. However, at update times, the separated parts must be combined, requiring an all-to-all agent communication in the correction step. Another DCL algorithm based on decoupling the propagation stage of an EKF CL using new intermediate variables is proposed in [21]. But here, unlike [22], at the update stage, each robot can locally reproduce the updated pose estimate and covariance of the centralized EKF after receiving an

update-message only from the robot that has made the relative measurement. Subsequently, [14] and [33] present DCL strategies using the MAP estimation procedure. In [14], computations of a centralized MAP CL algorithm are distributed among all the team members. In [33], the amount of data required to be passed between mobile agents to obtain the benefits of cooperative trajectory estimation locally is reduced by letting each agent treat the others as moving beacons whose estimate of positions is only required at communication/measurement times. The aforementioned techniques all assume that communication messages are delivered, as prescribed, perfectly, all the time. A DCL approach equivalent to a centralized CL, when possible, which handles both limited communication ranges and time-varying communication graphs, is proposed in [32]. This technique uses an information transfer scheme wherein each agent broadcasts all its locally available information to every agent within its communication radius at each time step. The broadcast information of each agent includes the past and present measurements as well as past measurements previously received from other agents. The main drawback of this method is its high communication and memory cost, which may not be affordable in applications with limited communication bandwidth and storage resources.

THE INTERIM MASTER DCL ALGORITHM: A TIGHTLY COUPLED DCL STRATEGY BASED ON KALMAN FILTER DECOUPLING

Because of its recursive and simple structure, the EKF is a very popular estimation strategy. However, as discussed in the “Cooperative Localization via the Extended Kalman Filter” section, a naive decentralized implementation of EKF requires an all-to-all communication at every time step of the algorithm. This section describes how, by exploiting a special pattern in the propagation estimation equations, tightly coupled *exact* decentralized implementations of EKF for CL with reduced communication workload per agent can be obtained. Here, “exact” means that if these decentralized implementations are initialized the same as a centralized EKF, then they produce the same state estimate and the associated state-error covariance of the centralized filter. The focus in this section is on the algorithm of [22] and a detailed treatment of the interim master DCL (IMDCL) algorithm of [21].

The algorithm of [22] and the IMDCL algorithm are developed based on the observation that, in localization problems, normally, the interest is in the *explicit* value of the pose estimate and the error covariance associated with it, while cross-covariance terms are only required in the update equations. Such an observation promoted the proposal of an *implicit* tracking of cross-covariance terms by splitting them into intermediate variables that can be propagated locally by the agents. Then, cross-covariance terms can be recovered by putting together these intermediate

variables at any update incidence. Let the last measurement update be in time step k and assume that for m subsequent and consecutive steps no relative measurement incidence takes place among the team members, that is, no intermediate measurement update is conducted in this time interval. In such a scenario, the propagated cross-covariance terms for these m consecutive steps are

$$\mathbf{P}_{ij}^-(k+l) = \mathbf{F}^i(k+l-1) \cdots \mathbf{F}^i(k) \mathbf{P}_{ij}^+(k) \mathbf{F}^j(k)^\top \cdots \mathbf{F}^j(k+l-1)^\top, \quad (8)$$

for $l \in \{1, \dots, m\}$, $i \in \mathcal{V}$, and $j \in \mathcal{V} \setminus \{i\}$. That is, at each time step after k , the propagated cross-covariance term is obtained by recursively multiplying its previous value by the Jacobian of the system function of agent i on the left and by the transpose of the Jacobian of the system function of agent j at that time step on the right. Based on this observation, [22] decomposed the last updated cross-covariance term $\mathbf{P}_{ij}^+(k)$ between any agent i and any other agent j of the team into two parts (for example, using the singular-value decomposition technique). Then, agent i propagated the left portion, and agent j propagated the right portion. Note that, as long as there is no relative measurement among team members, each agent can propagate its portion of the cross-covariance term locally without needing to communicate with others. Such a decomposition in [22] led to a fully decentralized estimation algorithm during the propagation cycle. However, in the update stage, all the agents needed to communicate with one another to put together the split cross-covariance terms and proceed with the update stage. The approach to obtain the IMDCL, which is outlined below, is also based on using intermediate variables to decouple the cross-covariance propagation equations. However, this alternative decomposition allows every agent to update its pose estimate and its associated covariance in a centralized-equivalent manner, using merely a scalable communication message that is received from the team member that takes the relative measurement. As such, the IMDCL algorithm removes the necessity of an all-to-all communication in the update stage and replaces it with propagating a constant-size communication message that holds the crucial piece of information needed in the update stage.

In particular, the IMDCL algorithm is developed based on the observation that $\mathbf{P}_{ij}^-(k+l)$ in (8) is composed of the following three parts: 1) $\mathbf{F}^i(k+l-1) \cdots \mathbf{F}^i(k)$, which is local to agent i , 2) the $\mathbf{P}_{ij}^+(k)$, which does not change unless there is relative measurement among the team members, and 3) $\mathbf{F}^j(k)^\top \cdots \mathbf{F}^j(k+l-1)^\top$, which is local to agent j . Motivated by how these three parts evolve in time, the propagated cross covariances (S1c) are written as

$$\mathbf{P}_{ij}^-(k+1) = \Phi^i(k+1) \mathbf{\Pi}_{ij}(k) \Phi^j(k+1)^\top, \quad k \in \{0, 1, 2, \dots\}, \quad (9)$$

where, for $i \in \mathcal{V}$,

$$\Phi^i(0) = \mathbf{I}_{n^i}, \quad \Phi^i(k+1) = \mathbf{F}^i(k) \Phi^i(k), \quad k \in \{0, 1, 2, \dots\}. \quad (10)$$

It is interesting to notice the resemblance of (10) and the transition matrix for discrete-time systems. The variable $\mathbf{\Pi}_{ij} \in \mathbb{R}^{n^i \times n^j}$, for $i, j \in \mathcal{V}$ and $i \neq j$ in (9) is also time varying and initialized at $\mathbf{\Pi}_{ij}(0) = \mathbf{0}_{n^i \times n^j}$. When there is no relative measurement at time $k+1$, (9) results in $\mathbf{\Pi}_{ij}(k+1) = \mathbf{\Pi}_{ij}(k)$. However, when there is a relative measurement among the team members, $\mathbf{\Pi}_{ij}$ must be updated in a manner such that $\mathbf{P}_{ij}^-(k+2) = \Phi^i(k+2) \mathbf{\Pi}_{ij}(k+1) \Phi^j(k+2)^\top$. To this end, notice that the update equations (S2) and (S4) of the centralized CL algorithm can be rewritten by replacing the cross-covariance terms by (9). (Recall that in the update stage, the assumption is that robot a has taken measurement from robot b)

$$\begin{aligned} \mathbf{S}_{ab} &= \mathbf{R}^a + \tilde{\mathbf{H}}_a \mathbf{P}^{a-}(k+1) \tilde{\mathbf{H}}_a^\top + \tilde{\mathbf{H}}_b \mathbf{P}^{b-}(k+1) \tilde{\mathbf{H}}_b^\top \\ &\quad - \underbrace{\tilde{\mathbf{H}}_a \Phi^a(k+1) \mathbf{\Pi}_{ab}(k) \Phi^b(k+1)^\top \tilde{\mathbf{H}}_b^\top}_{\mathbf{P}_{ab}^-(k+1)} \\ &\quad - \underbrace{\tilde{\mathbf{H}}_b \Phi^b(k+1) \mathbf{\Pi}_{ba}(k) \Phi^a(k+1)^\top \tilde{\mathbf{H}}_a^\top}_{\mathbf{P}_{ba}^-(k+1)}, \end{aligned} \quad (11)$$

and the Kalman gain is

$$\mathbf{K}_i = \Phi^i(k+1) \Gamma_i \mathbf{S}_{ab}^{-\frac{1}{2}}, \quad i \in \mathcal{V},$$

where

$$\Gamma_i = (\mathbf{\Pi}_{ib}(k) \Phi^{b\top} \tilde{\mathbf{H}}_b^\top - \mathbf{\Pi}_{ia}(k) \Phi^{a\top} \tilde{\mathbf{H}}_a^\top) \mathbf{S}_{ab}^{-\frac{1}{2}}, \quad i \in \mathcal{V} \setminus \{a, b\}, \quad (12a)$$

$$\Gamma_a = (\mathbf{\Pi}_{ab}(k) \Phi^{b\top} \tilde{\mathbf{H}}_b^\top - (\Phi^a)^{-1} \mathbf{P}^{a-} \tilde{\mathbf{H}}_a^\top) \mathbf{S}_{ab}^{-\frac{1}{2}}, \quad (12b)$$

$$\Gamma_b = ((\Phi^b)^{-1} \mathbf{P}^{b-} \tilde{\mathbf{H}}_b^\top - \mathbf{\Pi}_{ba}(k) \Phi^{a\top} \tilde{\mathbf{H}}_a^\top) \mathbf{S}_{ab}^{-\frac{1}{2}}. \quad (12c)$$

Generally, $\mathbf{F}^i(k)$ is invertible for all $k \geq 0$ and $i \in \mathcal{V}$. Therefore, $\Phi^i(k)$, for all $k \geq 0$ and $i \in \mathcal{V}$, is invertible.

Next, for $i \neq j$ and $i, j \in \mathcal{V}$, the cross-covariance terms (S3c) are written as

$$\begin{aligned} \mathbf{P}_{ij}^+(k+1) &= \mathbf{P}_{ij}^-(k+1) - \mathbf{K}_i \mathbf{S}_{ab} \mathbf{K}_j^\top \\ &= \Phi^i(k+1) \mathbf{\Pi}_{ij}(k) \Phi^j(k+1)^\top \\ &\quad - (\Phi^i(k+1) \Gamma_i \mathbf{S}_{ab}^{-\frac{1}{2}}) \mathbf{S}_{ab} (\Phi^j(k+1) \Gamma_j \mathbf{S}_{ab}^{-\frac{1}{2}})^\top \\ &= \Phi^i(k+1) (\mathbf{\Pi}_{ij}(k) - \Gamma_i \Gamma_j^\top) \Phi^j(k+1)^\top. \end{aligned}$$

Let $\mathbf{\Pi}_{ij}(k+1) = \mathbf{\Pi}_{ij}(k) - \Gamma_i \Gamma_j^\top$. Then, the cross-covariance update (S3c) can be rewritten as

$$\mathbf{P}_{ij}^+(k+1) = \Phi^i(k+1) \mathbf{\Pi}_{ij}(k+1) \Phi^j(k+1)^\top. \quad (13)$$

Therefore, at time $k+2$, the propagated cross-covariances terms for $i \neq j$ and $i, j \in \mathcal{V}$ are

$$\begin{aligned} \mathbf{P}_{ij}^-(k+2) &= \mathbf{F}^i(k+1) \mathbf{P}_{ij}^+(k+1) \mathbf{F}^j(k+1)^\top \\ &= \mathbf{F}^i(k+1) \Phi^i(k+1) \mathbf{\Pi}_{ij}(k+1) \Phi^j(k+1)^\top \mathbf{F}^j(k+1)^\top \\ &= \Phi^i(k+2) \mathbf{\Pi}_{ij}(k+1) \Phi^j(k+2)^\top. \end{aligned}$$

In short, the propagated and the updated cross-covariance terms of the centralized EKF CL can be written as, respectively, (9) and (13) for all $k \in \{0, 1, \dots\}$ where the $\Phi^i(k)$ of each robot evolves according to (10) and

$$\mathbf{\Pi}_{ij}(k+1) = \begin{cases} \mathbf{\Pi}_{ij}(k), & \text{no relative measurement at } k+1, \\ \mathbf{\Pi}_{ij}(k) - \Gamma_i \Gamma_j^\top, & \text{otherwise,} \end{cases} \quad (14)$$

for $i, j \in \mathcal{V}$ and $i \neq j$.

ALGORITHM 2 Interim master decentralized cooperative localization (IMDCL).

Require: Initialization ($k = 0$): Every agent $i \in \mathcal{V}$ initializes its filter at

$$\hat{\mathbf{x}}^{i+}(0) \in \mathbb{R}^{n'}, \mathbf{P}^{i+}(0) \in \mathbb{M}_{n'}, \Phi^i(0) = \mathbf{I}_{n'}, \Pi_{jl}^i(0) = \mathbf{0}_{n' \times n'}, j \in \mathcal{V} \setminus \{N\}, l \in \{j+1, \dots, N\}. \quad (\text{S5})$$

Iteration k

1: Propagation: Every agent $i \in \mathcal{V}$ propagates the variables below

$$\hat{\mathbf{x}}^{i+}(k+1) = \mathbf{f}^i(\hat{\mathbf{x}}^{i+}(k), \mathbf{u}^i(k)), \mathbf{P}^{i+}(k+1) = \mathbf{F}^i(k) \mathbf{P}^{i+}(k) \mathbf{F}^{i+}(k)^T + \mathbf{G}^i(k) \mathbf{Q}^i(k) \mathbf{G}^{i+}(k)^T, \Phi^i(k+1) = \mathbf{F}^i(k) \Phi^i(k). \quad (\text{S6})$$

2: Update: while there are no relative measurements in the network, every agent $i \in \mathcal{V}$ updates its variables as

$$\hat{\mathbf{x}}^{i+}(k+1) = \hat{\mathbf{x}}^{i-}(k+1), \mathbf{P}^{i+}(k+1) = \mathbf{P}^{i-}(k+1), \Pi_{jl}^i(k+1) = \Pi_{jl}^i(k), j \in \mathcal{V} \setminus \{N\}, l \in \{j+1, \dots, N\}. \quad (\text{S7})$$

If there is an agent a that makes a measurement with respect to another agent b , then agent a is declared as the interim master and acquires the following information from agent b

$$\text{landmark-message} = (\hat{\mathbf{x}}^{b-}(k+1), \Phi^b(k+1), \mathbf{P}^{b-}(k+1)). \quad (\text{S8})$$

Agent a makes the following calculations upon receiving the landmark-message

$$\mathbf{r}^a = \mathbf{z}_{ab} - \mathbf{h}_{ab}(\hat{\mathbf{x}}^{a-}, \hat{\mathbf{x}}^{b-}), \quad (\text{S9a})$$

$$\mathbf{S}_{ab} = \mathbf{R}^a + \tilde{\mathbf{H}}_a \mathbf{P}^{a-} \tilde{\mathbf{H}}_a^T + \tilde{\mathbf{H}}_b \mathbf{P}^{b-} \tilde{\mathbf{H}}_b^T - \tilde{\mathbf{H}}_a \Phi^a \Pi_{ab}^a \Phi^{bT} \tilde{\mathbf{H}}_b^T - \tilde{\mathbf{H}}_b \Phi^b \Pi_{ba}^b \Phi^{aT} \tilde{\mathbf{H}}_a^T, \quad (\text{S9b})$$

$$\Gamma_a = ((\Phi^a)^{-1} \Phi^a \Pi_{ab}^a \Phi^{bT} \tilde{\mathbf{H}}_b^T - (\Phi^a)^{-1} \mathbf{P}^{a-} \tilde{\mathbf{H}}_a^T) \mathbf{S}_{ab}^{-\frac{1}{2}}, \Gamma_b = ((\Phi^b)^{-1} \mathbf{P}^{b-} \tilde{\mathbf{H}}_b^T - \Pi_{ba}^b \Phi^{aT} \tilde{\mathbf{H}}_a^T) \mathbf{S}_{ab}^{-\frac{1}{2}}, \quad (\text{S9c})$$

where $\tilde{\mathbf{H}}_a(k+1) = \tilde{\mathbf{H}}_a(\hat{\mathbf{x}}^{a-}, \hat{\mathbf{x}}^{b-})$ and $\tilde{\mathbf{H}}_b(k+1) = \tilde{\mathbf{H}}_b(\hat{\mathbf{x}}^{a-}, \hat{\mathbf{x}}^{b-})$ are obtained using (6). The interim master passes the following data, either directly or indirectly (by message passing), to the rest of the agents in the network

$$\text{update-message} = (a, b, \bar{\mathbf{r}}^a, \Gamma_a, \Gamma_b, \Phi^{bT} \tilde{\mathbf{H}}_b^T \mathbf{S}_{ab}^{-\frac{1}{2}}, \Phi^{aT} \tilde{\mathbf{H}}_a^T \mathbf{S}_{ab}^{-\frac{1}{2}}). \quad (\text{S10})$$

Every agent $i \in \mathcal{V}$, upon receiving the update-message, first calculates, for all $j \in \mathcal{V} \setminus \{a, b\}$, using information obtained at k

$$\Gamma_j = \Pi_{jb}^i \Phi^{bT} \tilde{\mathbf{H}}_b^T \mathbf{S}_{ab}^{-\frac{1}{2}} - \Pi_{ja}^i \Phi^{aT} \tilde{\mathbf{H}}_a^T \mathbf{S}_{ab}^{-\frac{1}{2}}, \quad (\text{S11})$$

and then updates the following variables

$$\hat{\mathbf{x}}^{i+}(k+1) = \hat{\mathbf{x}}^{i-}(k+1) + \Phi^i(k+1) \Gamma_i \bar{\mathbf{r}}^a, \quad (\text{S12a})$$

$$\mathbf{P}^{i+}(k+1) = \mathbf{P}^{i-}(k+1) - \Phi^i(k+1) \Gamma_i \Gamma_i^T \Phi^i(k+1)^T, \quad (\text{S12b})$$

$$\Pi_{jl}^i(k+1) = \Pi_{jl}^i(k) - \Gamma_j \Gamma_j^T, j \in \mathcal{V} \setminus \{N\}, l \in \{j+1, \dots, N\}. \quad (\text{S12c})$$

3: $k \leftarrow k+1$

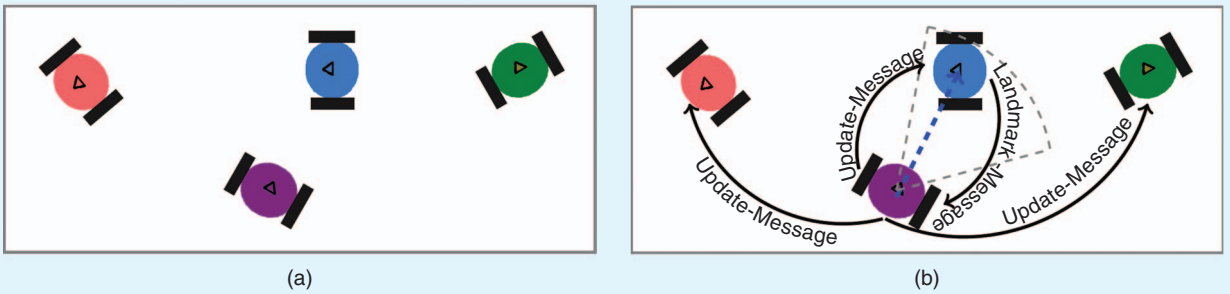


FIGURE 6 The in-network information flow of the interim master decentralized cooperative localization (IMDCL) algorithm. In the IMDCL algorithm, communication is only needed in the update stage when the team members use a robot-to-robot relative measurement feedback to correct their pose estimation. Here, it is assumed that all the team members are in the communication range of the interim master robot. (a) The propagation stage for all $k \in \{0, 1, 2, \dots\}$. (b) The update stage.

Next, notice that, for $i \in \mathcal{V}$, the updated state estimate and covariance matrix in the new variables can be written as

$$\begin{aligned} \hat{\mathbf{x}}^{i+}(k+1) &= \hat{\mathbf{x}}^{i-}(k+1) + \Phi^i(k+1) \Gamma_i \bar{\mathbf{r}}^a, \\ \mathbf{P}^{i+}(k+1) &= \mathbf{P}^{i-}(k+1) - \Phi^i(k+1) \Gamma_i \Gamma_i^T \Phi^i(k+1)^T, \end{aligned} \quad (15)$$

where $\bar{\mathbf{r}}^a = \mathbf{S}_{ab}^{-\frac{1}{2}} \mathbf{r}^a$.

Using the alternative representations (9), (13), and (15) of the EKF CL, the decentralized implementation IMDCL is given in Algorithm 2. The IMDCL algorithm is developed by keeping a local copy of Π_{jl}^i at each agent $i \in \mathcal{V}$, that is, Π_{jl}^i for all $j \in \mathcal{V} \setminus \{N\}$ and $l \in \{j+1, \dots, N\}$. Here, because of the symmetry of the covariance matrix only, for example, the upper triangular part of this matrix is needed to be stored

**Because of its recursive and simple structure,
the EKF is a very popular estimation strategy.**

and to be evolved. For example, for a group of $N = 4$ robots, every agent maintains a copy of $\{\Pi_{12}^i, \Pi_{13}^i, \Pi_{14}^i, \Pi_{23}^i, \Pi_{24}^i, \Pi_{34}^i\}$. During the algorithm implementation, it is assumed that if Π_{jl}^i is not explicitly maintained by agent i , the agent substitutes the value of $(\Pi_{ij}^i)^\top$ for it.

In IMDCL, every agent $i \in \mathcal{V}$ initializes its own state-estimate $\hat{\mathbf{x}}^{i+}(0)$, the error covariance matrix $\mathbf{P}^{i+}(0)$, $\Phi^i(0) = \mathbf{I}_{n^i}$, and its local copies $\Pi_{jl}^i(0) = \mathbf{0}_{n^i \times n^l}$, for $j \in \mathcal{V} \setminus \{N\}$ and $l \in \{j+1, \dots, N\}$; see (S5). At the propagation stage, every agent evolves its local state estimate, error covariance and Φ^i , according to, respectively, (S1a), (S1b), and (10); see (S6). At every time step, when there is no exteroceptive measurement in the team, the local updated state estimates and error covariance matrices are replaced by their respective propagated counterparts, while each Π_{jl}^i , to respect (14), is kept unchanged; see (S7). When there is a robot-to-robot measurement, examining (6), (5a), (11), (12b), and (12c) shows that agent a , the robot that made the relative measurement, can calculate these terms using its local Π_{jl}^i and acquiring $\hat{\mathbf{x}}^{b-}(k+1) \in \mathbb{R}^{n^b}$, $\Phi^b(k+1) \in \mathbb{R}^{n^b \times n^b}$, and $\mathbf{P}^{b-}(k+1) \in \mathbb{M}_{n^b}$; see (S8) and (S9). Then, agent a can assume the role of the interim master and issue the update terms for other agents in the team; see (S10). Using this update-message and their local variables, then each agent $i \in \mathcal{V}$ can compute (12a) and obtain its local state updates of (15) and (14); see (S12). Figure 6 demonstrates the information flow direction between agents while implementing the IMDCL algorithm.

The inclusion of absolute measurements in the IMDCL is straightforward. The agent making an absolute measurement

is an interim master that can calculate the *update-message* using only its own data and then broadcast it to the team. Next, observe that the IMDCL algorithm is robust to permanent agent dropouts from the network. The operation only suffers from a processing cost until all agents become aware of the dropout. Also, notice that an external authority, for example, a search-and-rescue chief, who needs to obtain the location of any agent, can obtain this location update at any rate (s) he wishes to by communicating with that agent. This reduces the communication cost of the external authority.

The IMDCL algorithm works under the assumption that the message from the agent taking the relative measurement, the interim master, reaches the entire team. Any communication failure results in a mismatch between the local copies of Π_{ij} at the agents receiving and missing the communication message. Readers are referred to [34] for a variation of IMDCL that is robust to intermittent communication message dropouts. Such guarantees in [34] are provided by replacing the fully decentralized implementation with a partial decentralization where a shared memory stores and updates the Π_{ij} , $j \in \mathcal{V} \setminus \{N\}$ and $l \in \{j+1, \dots, N\}$.

Complexity Analysis

For the sake of an objective performance evaluation, a study of the computational complexity, the memory usage, as well as communication cost per agent per time step of the IMDCL algorithm in terms of the size of the mobile-agent team N is provided next. At the propagation state of the IMDCL algorithm, the computations per agent are independent of the size of the team. However, at the update

TABLE 1 The complexity analysis per agent of the interim master decentralized cooperative localization (IMDCL) algorithm compared to that of the trivial decentralized implementation of extended Kalman filter for cooperative localization (denoted by T-DCL).

	Computation		Storage		Broadcast*		Message Size		Connectivity	
Algorithm	IMDCL	T-DCL	IMDCL	T-DCL	IMDCL	T-DCL	IMDCL	T-DCL	IMDCL	T-DCL
Propagation	$O(1)$	$O(N^2)$	$O(N^2)$	$O(N^2)$	0	$O(N)$	0	$O(1)$	None	Strongly connected digraph
Update per N_z relative measurement	$O(N_z \times N^2)$	$O(N_z \times N^2)$	$O(N^2)$	$O(N^2)$	$O(N_z)$	$O(N_z)$	$O(1)$	$O(1)$	Interim master can reach all the agents	
Overall worst case	$O(N^4)$	$O(N^4)$	$O(N^2)$	$O(N^2)$	$O(N^2)$	$O(N^2)$	$O(1)$	$O(1)$		

*Broadcast cost is for multihop communication. If the communication range is unbounded, the broadcast cost per agent is $O(\max\{N_b, N_a\})$ with the worst cost of $O(N)$.

stage, for each measurement update, the computation of every agent is of order $N(N-1)/2$ due to (S18c). Because multiple relative measurements are processed sequentially, the computational cost per agent at the completion of any update stage depends on the number of the relative measurements in the team, henceforth denoted by N_z . Then, the computational cost per agent is $O(N_z \times N^2)$, implying a computational complexity of order $O(N^4)$ for the worst case where all the agents take relative measurement with respect to all the other agents in the team, that is, $N_z = N(N-1)$. The memory cost per agent is of order $O(N^2)$, which, due to the recursive nature of the IMDCL algorithm, is independent of N_z . This cost is caused by the initialization (S5) and update equation (S18c), which are of order $N(N-1)/2$.

The analysis of the communication cost below is carried out for a multihop communication strategy. The IMDCL requires communication only in the update stage, where landmark robots should broadcast their landmark-message to their respective master, and every agent should rebroad-

cast any update-message it receives. Let N_r be the number of agents that have made a relative measurement at the current time, that is, $N_r \leq N$ is the number of current sequential interim masters. These robots should announce their identity and the number of their landmark robots to the entire team for sequential update cuing purpose, incurring a communication cost of order N_r per robot. Next, the team will proceed by sequentially processing the relative measurements. Every agent can be a landmark of $N_a \leq N_r$ agents and/or a master of $N_b \leq (N-1)$ agents. The updating procedure starts by a landmark robot sending its landmark-message to its active interim master, resulting in a total communication cost of $O(N_a)$ per landmark robot at the end of update stage. Every active interim master passes an update-message to the entire team, resulting in a total communication cost of $O(N_b)$ per robot. Because there are N_r masters, at the end of the update stage, every robot incurs a communication cost of $O(N_r \times N_b)$ to pass the update-messages. Because $N_a, N_r < N_r \times N_b \leq N_z$, the total communication cost at the end of the update stage $O(N_z)$

ALGORITHM 3 Alternative interim master decentralized cooperative localization (IMDCL) with a larger communication message size in favor of lower computation and storage cost per agent.

Require: Initialization ($k = 0$): Every agent $i \in V$ initializes its filter at

$$\hat{\mathbf{x}}^{i+}(0) \in \mathbb{R}^{n^i}, \mathbf{P}^{i+}(0) \in \mathbb{M}_{n^i}, \Phi^i(0) = \mathbf{I}_{n^i}, \Pi_{ij}^i(0) = \mathbf{0}_{n^i \times n^j} \in \mathcal{V}\{i\}. \quad (\text{S13})$$

Iteration k

1: Propagation: Every agent $i \in V$ propagates the variables below

$$\hat{\mathbf{x}}^{i-}(k+1) = \mathbf{f}^i(\hat{\mathbf{x}}^{i+}(k), \mathbf{u}^i(k)), \mathbf{P}^{i-}(k+1) = \mathbf{F}^i(k) \mathbf{P}^{i+}(k) \mathbf{F}^{i\top}(k) + \mathbf{G}^i(k) \mathbf{Q}^i(k) \mathbf{G}^{i\top}(k), \Phi^i(k+1) = \mathbf{F}^i(k) \Phi^i(k). \quad (\text{S14})$$

2: Update: while there are no relative measurements in the network, every agent $i \in V$ updates its variables as

$$\hat{\mathbf{x}}^{i+}(k+1) = \hat{\mathbf{x}}^{i-}(k+1), \mathbf{P}^{i+}(k+1) = \mathbf{P}^{i-}(k+1), \Pi_{ij}^i(k+1) = \Pi_{ij}^i(k), j \in V \setminus \{i\}.$$

If there is an agent a that makes a measurement with respect to another agent b , then agent a is declared as the interim master and acquires the following information from agent b

$$\text{landmark-message} = (\hat{\mathbf{x}}^{b-}(k+1), \Phi^b(k+1), \mathbf{P}^{b-}(k+1), \Pi_{bj}^b(k) \text{ where } j \in V \setminus \{a, b\}). \quad (\text{S15})$$

Agent a makes the following calculations upon receiving the landmark-message

$$\mathbf{r}^a = \mathbf{z}_{ab} - \mathbf{h}_{ab}(\hat{\mathbf{x}}^{a-}, \hat{\mathbf{x}}^{b-}), \quad (\text{S16a})$$

$$\mathbf{S}_{ab} = \mathbf{R}^a + \tilde{\mathbf{H}}_a \mathbf{P}^{a-} \tilde{\mathbf{H}}_a^\top + \tilde{\mathbf{H}}_b \mathbf{P}^{b-} \tilde{\mathbf{H}}_b^\top - \tilde{\mathbf{H}}_a \Phi^a \Pi_{ab}^a \Phi^{b\top} \tilde{\mathbf{H}}_b^\top - \tilde{\mathbf{H}}_b \Phi^b (\Pi_{ab}^a)^\top \Phi^{a\top} \tilde{\mathbf{H}}_a^\top, \quad (\text{S16b})$$

$$\Gamma_a = ((\Phi^a)^{-1} \Phi^a \Pi_{ab}^a \Phi^{b\top} \tilde{\mathbf{H}}_b^\top - (\Phi^a)^{-1} \mathbf{P}^{a-} \tilde{\mathbf{H}}_a^\top) \mathbf{S}_{ab}^{-\frac{1}{2}}, \Gamma_b = ((\Phi^b)^{-1} \mathbf{P}^{b-} \tilde{\mathbf{H}}_b^\top - (\Pi_{ab}^a)^\top \Phi^{a\top} \tilde{\mathbf{H}}_a^\top) \mathbf{S}_{ab}^{-\frac{1}{2}}, \quad (\text{S16c})$$

$$\Gamma_j = (\Pi_{bj}^b)^\top \Phi^{b\top} \tilde{\mathbf{H}}_b^\top \mathbf{S}_{ab}^{-\frac{1}{2}} - (\Pi_{aj}^a)^\top \Phi^{a\top} \tilde{\mathbf{H}}_a^\top \mathbf{S}_{ab}^{-\frac{1}{2}}, j \in V \setminus \{a, b\}, \quad (\text{S16d})$$

where $\tilde{\mathbf{H}}_a(k+1) = \tilde{\mathbf{H}}_a(\hat{\mathbf{x}}^{a-}, \hat{\mathbf{x}}^{b-})$ and $\tilde{\mathbf{H}}_b(k+1) = \tilde{\mathbf{H}}_b(\hat{\mathbf{x}}^{a-}, \hat{\mathbf{x}}^{b-})$ are obtained using (6).

The interim master passes the following data, either directly or indirectly (by message passing), to the rest of the agents in the network

$$\text{update-message} = (a, b, \mathbf{r}^a, \Gamma_1, \dots, \Gamma_N). \quad (\text{S17})$$

Every agent $i \in V$, upon receiving the update-message, updates the following variables

$$\hat{\mathbf{x}}^{i+}(k+1) = \hat{\mathbf{x}}^{i-}(k+1) + \Phi^i(k+1) \mathbf{\Gamma}_i \mathbf{r}^a, \quad (\text{S18a})$$

$$\mathbf{P}^{i+}(k+1) = \mathbf{P}^{i-}(k+1) - \Phi^i(k+1) \mathbf{\Gamma}_i \mathbf{\Gamma}_i^\top \Phi^i(k+1)^\top, \quad (\text{S18b})$$

$$\Pi_{ij}^i(k+1) = \Pi_{ij}^i(k) - \mathbf{\Gamma}_i \mathbf{\Gamma}_j^\top, j \in V \setminus \{i\}. \quad (\text{S18c})$$

3: $k \leftarrow k+1$

per agent, implying a worst case broadcast cost of $O(N^2)$ per agent. If the communication range is unbounded, the broadcast cost per agent is $O(\max\{N_b, N_a\})$, with the worst-case cost of $O(N)$. The communication message size of each agent in both single or multiple relative measurements is independent of the group size N . As such, for the worst case scenario the communication message size is of $O(1)$.

The results of the analysis above are summarized in Table 1 and compared to those of a trivial decentralized implementation of the EKF for CL (denoted by T-DCL) in which every agent $i \in \mathcal{V}$ at the propagation stage computes (S1), using the broadcasted $\mathbf{F}^i(k)$ from every other team member $j \in \mathcal{V} \setminus \{i\}$, and at the update stage computes (S4) and (S3), using the broadcast $(a, b, \mathbf{r}^a, \mathbf{S}_{ab}, \hat{\mathbf{H}}_a, \hat{\mathbf{H}}_b, \mathbf{R}^a, \mathbf{P}^{a-}, \mathbf{P}^{b-})$ from agent a that has made relative measurement from agent b . Agent a calculates $\mathbf{S}_{ab}, \hat{\mathbf{H}}_a, \hat{\mathbf{H}}_b$ by requesting $(\hat{\mathbf{x}}^{b-}, \mathbf{P}^{b-})$ from agent b . Here, it is assumed that multiple measurements are processed sequentially and that the communication procedure is multihop. Although the overall cost of the T-DCL algorithm is comparable with the IMDCL algorithm, this implementation has a more stringent communication connectivity condition, requiring a *strongly connected digraph* topology (that is, all the nodes on the communication graph can be reached by every other node on the graph) at each time step, regardless of whether there is a relative measurement incidence in the team. As an example, notice that the communication graph of Figure 2 is not strongly connected and as such the T-DCL algorithm can not be implemented whereas the IMDCL algorithm can be. Recall that the IMDCL algorithm needs no communication at the propagation stage, and it only requires an existence of a spanning tree rooted at the agent making the relative measurement at the update stage. Finally, the IMDCL algorithm incurs less computational cost at the propagation stage.

Algorithm 3 presents an alternative IMDCL implementation where, instead of storing and evolving Π_{ij} s of the entire team, every agent only maintains the terms corresponding to its own cross covariances; see (S13) and (S14). For example, in a team of $N = 4$, robot 1 maintains $\{\Pi_{12}^1, \Pi_{13}^1, \Pi_{14}^1\}$ and robot 2 maintains $\{\Pi_{21}^2, \Pi_{23}^2, \Pi_{24}^2\}$. However, now the interim master a needs to acquire the Π_{ij} 's from the landmark robot b and calculate and broadcast $\Gamma_i, i \in \mathcal{V}$ to the entire team; see (S15), (S16), and (S18). In this alternative implementation, the processing and storage cost of every agent is reduced from $O(N^2)$ to $O(N)$, however the communication message size is increased from $O(1)$ to $O(N)$.

Tightly Coupled Versus Loosely Coupled DCL: A Numerical Comparison Study

The IMDCL falls under the tightly coupled DCL classification. Figure 7 demonstrates the positioning accuracy (time history of the root-mean square error plot for 50 Monte Carlo simulation runs) of this algorithm versus the loosely coupled EKF and covariance-intersection based algorithm of [25] in the following scenario. For this numerical study,

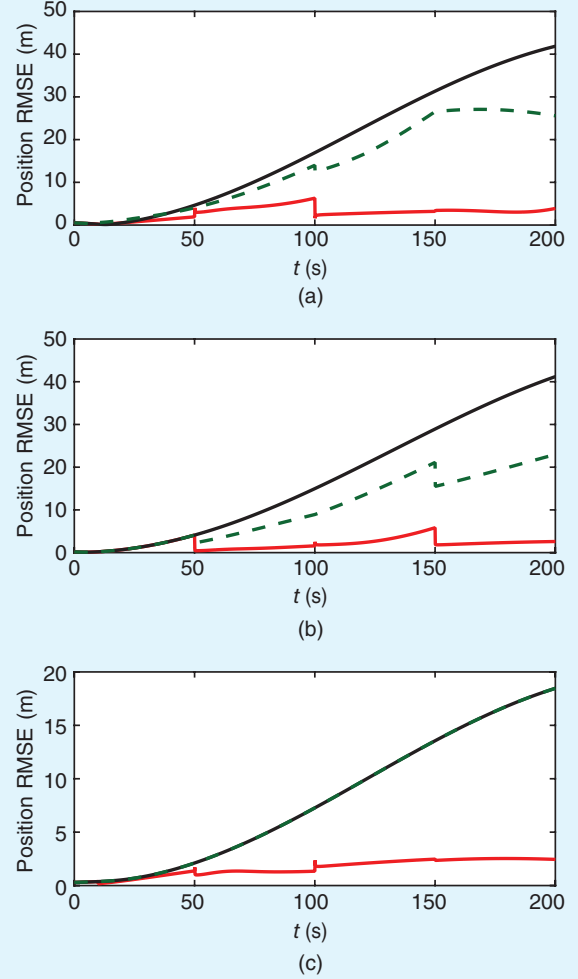


FIGURE 7 A comparison study between the positioning accuracy of three robots employing the interim master decentralized cooperative localization (IMDCL) algorithm (red plots), with that from the extended Kalman filter, covariance-intersection-based CL algorithm of [25] (dashed green plot). The curves in black show the positioning accuracy when the robots do not use any CL. As expected, by keeping an accurate account of the cross covariances, the IMDCL algorithm produces more accurate localization results than the algorithm of [25]. However, this higher accuracy comes with higher communication and processing cost per robot. Notice here that using algorithm of [25], robot 3 does not get to update its estimation equations. (a) Robot 1, (b) Robot 2, and (c) Robot 3.

consider again the three mobile robots employed in the numerical example of the section “Cooperative Localization via the Extended Kalman Filter” with motion as described in that section. The sensing scenario here is that, starting at $t = 10$ s, robot 3 takes persistent relative measurements alternating every 50 s from robot 1 to robot 2 and vice versa. As expected, the tightly coupled IMDCL algorithm produces more accurate position estimation results than those of the loosely coupled DCL algorithm of [25] (similar results can be observed for the heading estimation accuracy, which is omitted here for brevity).

The IMDCL algorithm works under the assumption that the message from the agent taking the relative measurement, the interim master, reaches the entire team.

In the algorithm of [25], every robot keeps an EKF estimate of its own pose. When a robot takes a relative pose measurement from another robot (here, also refer to the robot taking the relative measurement as the interim master), it acquires the current position estimate and the corresponding error covariance of the landmark robot to use along with its own current estimate and the current relative measurement to extract a new state estimate and the corresponding error covariance for the landmark robot. Next, the interim master robot transmits this new estimate to the landmark robot, which uses the covariance intersection method to fuse it, consistently, to its current pose estimate. It is interesting to notice that in this particular scenario, even though robot 3 has been taking all the relative measurements, it receives no benefit from such measurements because only the landmark robots are updating their estimates; see Figure 7. Even though the positioning accuracy of algorithm [25] is lower, it only requires $O(1)$ computational cost per agent as compared to the $O(N^2)$ cost of the IMDCL algorithm. However, it also requires more complicated calculations to perform covariance intersection fusion. If the communication range of each agent covers the entire team, then interestingly the communication cost of these two algorithms is the same because both use an $O(1)$ landmark- and update-messages. However, if the communication range is bounded, the loosely coupled algorithm of [25] offers a more flexible and cost-effective communication policy.

CONCLUSIONS

This article presented a brief review on CL as a strategy to increase the localization accuracy of a team of mobile agents with communication capabilities. This strategy relies on use of agent-to-agent relative measurements (no reliance on external features) as a feedback signal to *jointly estimate* the poses of the team members. In particular, this article discussed the challenges involved in designing DCL algorithms and presented a decentralized CL algorithm that is exactly equivalent to the centralized EKF algorithm of [22]. In this decentralized algorithm, the propagation stage is fully decoupled, that is, the propagation is a local calculation and no intranetwork communication is needed. The communication between agents is only required in the update stage when one agent makes a relative measurement with respect to another agent. The algorithm declares the agent made the measurement as interim master that

can, by using the data acquired from the landmark agent, calculate and deliver by broadcast the update terms for the rest of the team. Future extensions of this work include handling message dropouts and asynchronous measurement updates.

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